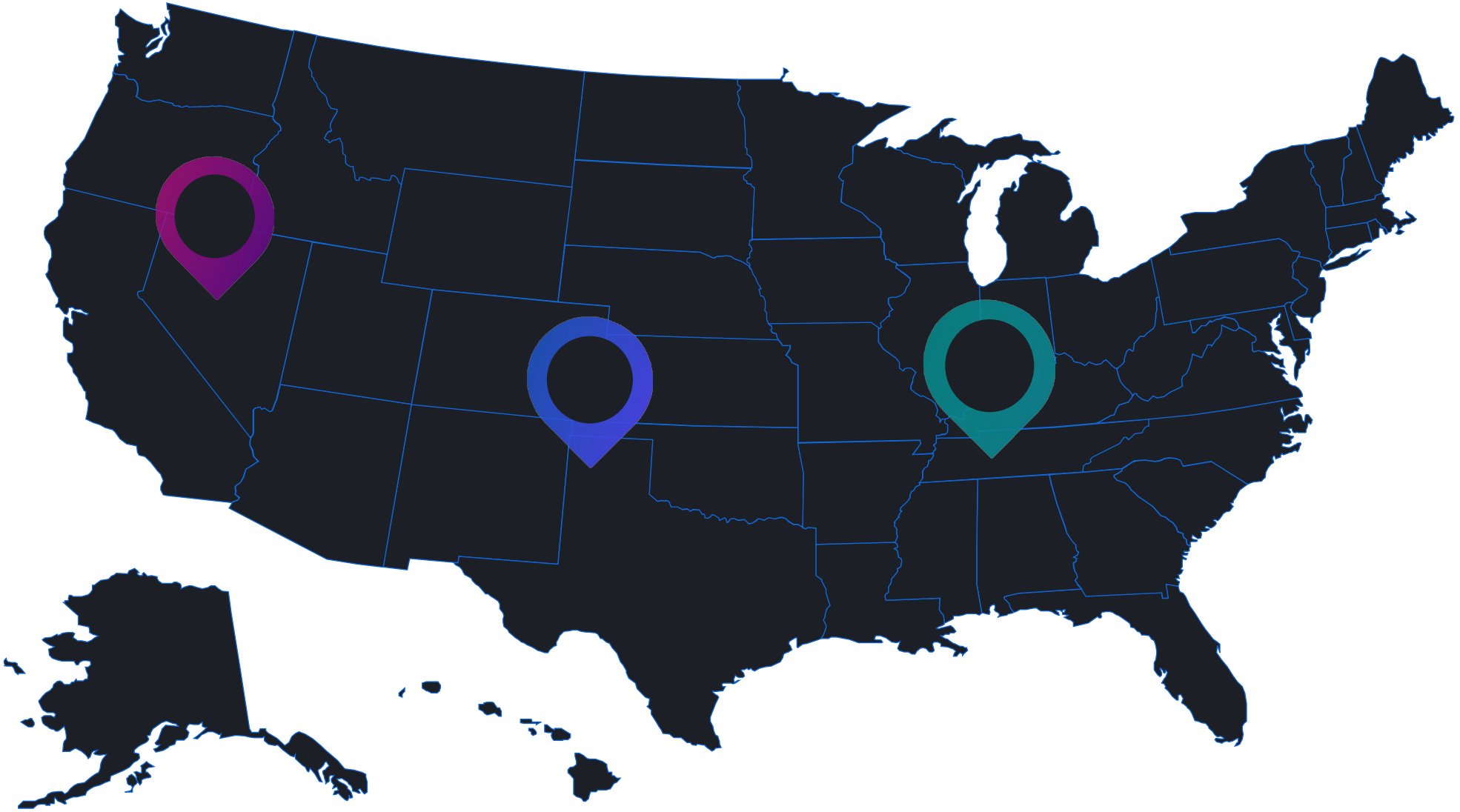
Public Transportation System



Abstract

- Transportation has become one of the biggest problems facing people in our city.

- An Intelligent Transportation System (ITS) is an advanced application that enables users to be better informed about transports.

- It makes transportation networks safer, more coordinated and smarter.

- In this project we focus on public transportation vehicles - such as buses.

Introduction

Main problems

- Transportation is an integral part of our lives yet it is filled with problems and inefficiencies.

- One of the most serious problems is people gathering while waiting for transportation.

Main problems

- Another problem is that some drivers don’t commit with the trips real time.

- This problem has more than one effect :
  - Some people will wait a lot in the street.
  - This problem will affect other trips time.

- Some people don’t know which transportation they should take to reach their destination.

System objectives

- Allow more smarter and intelligent transportations in the city.
- Grants citizens with an access to information about buses.
- Admins monitor all buses and trips.
- Automatic vehicle location system.

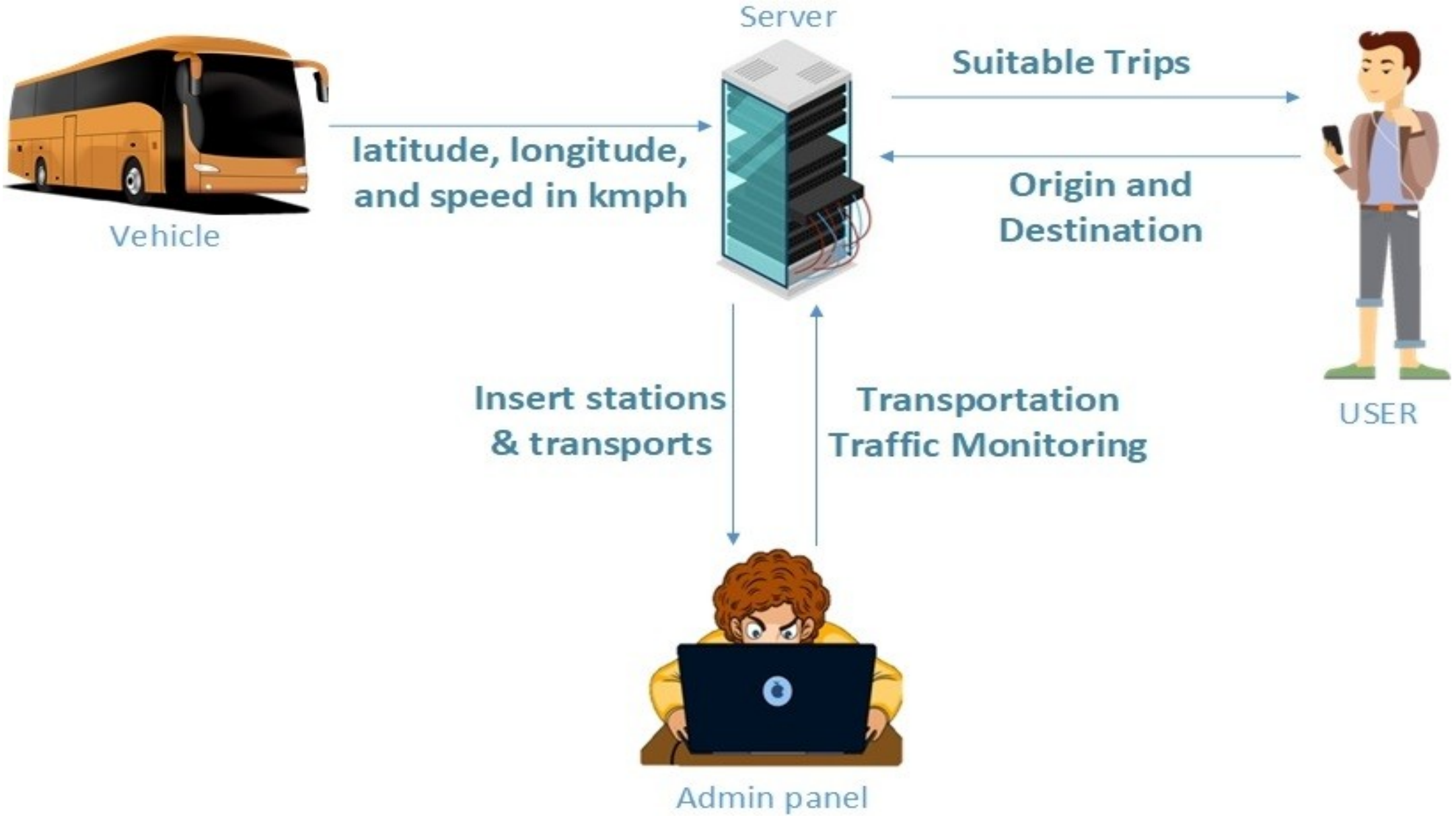
Benefits

- It improves transportation in the city.
- This system can save time to citizens and make them more comfortable.
- It helps people to deal with transportation easily.
- Reduces passenger congestion while waiting.

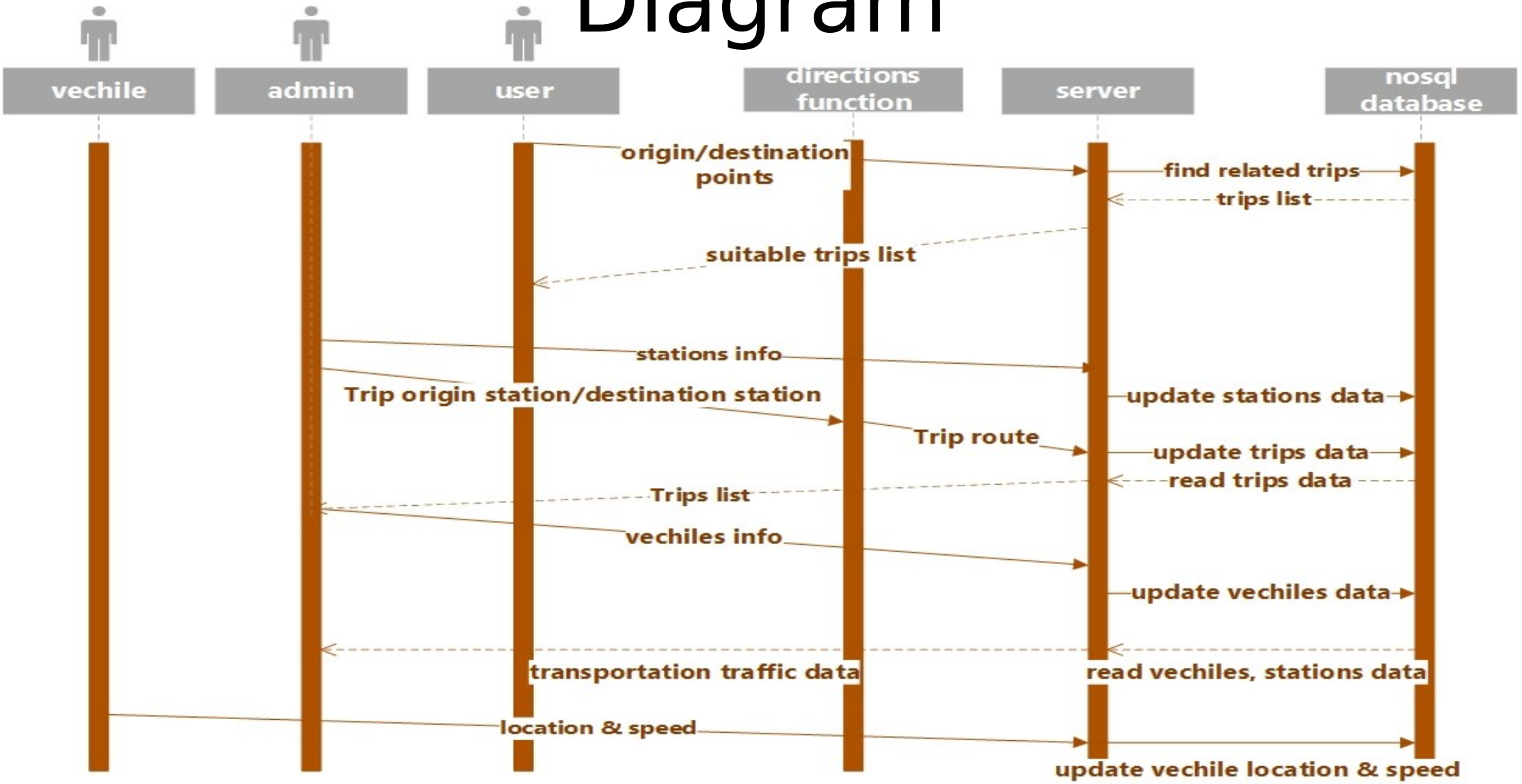
Main parts

- Admin Panel for monitoring the whole system.
- Mobile Application for users to show appropriate trips.
- Hardware part to access vehicles location.
- Cloud functions.

High level architecture



Sequence Diagram



Over View

Over view of how the user choose trip and bus

Trips search algorithm

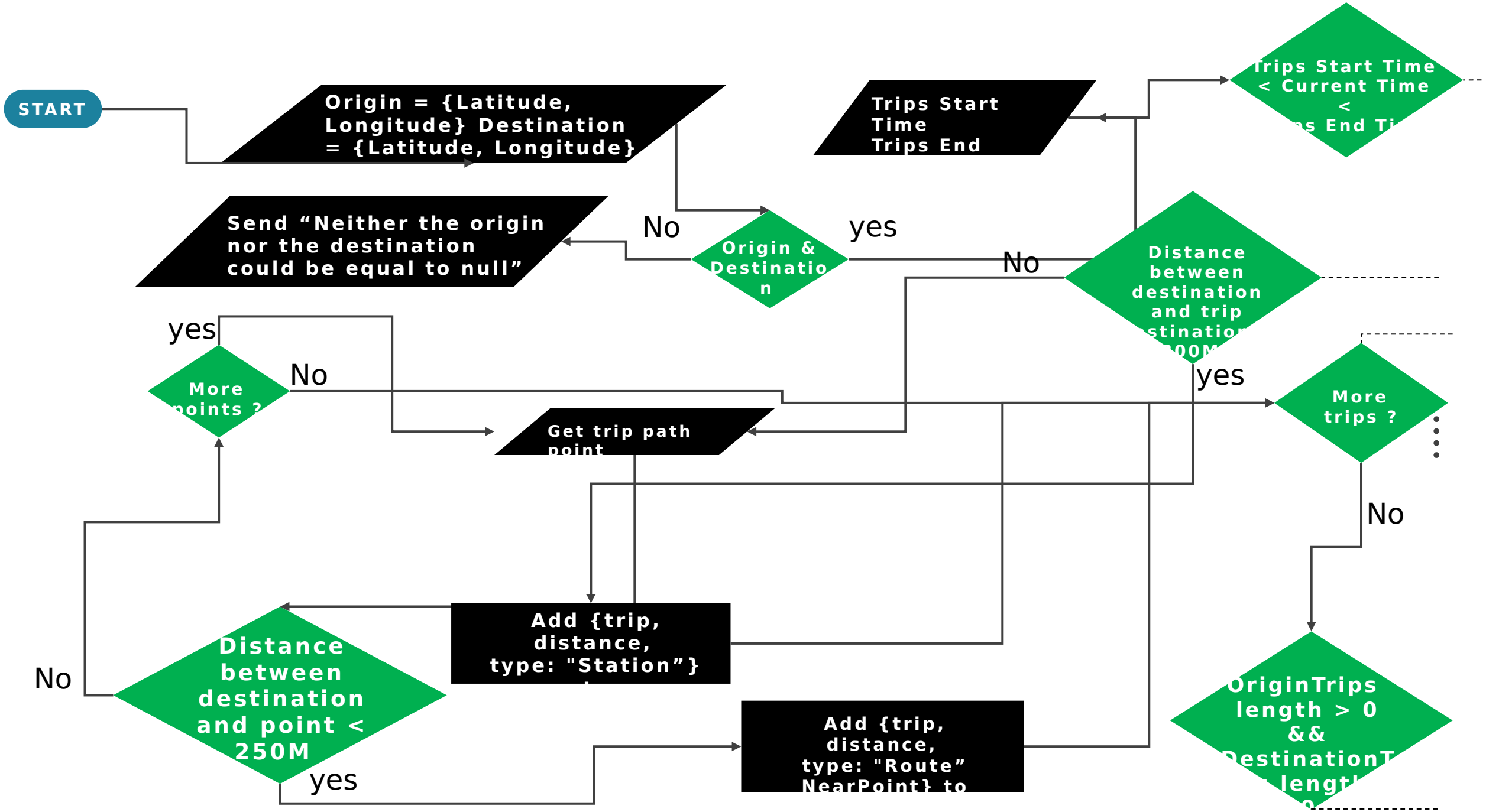
Trips search algorithm

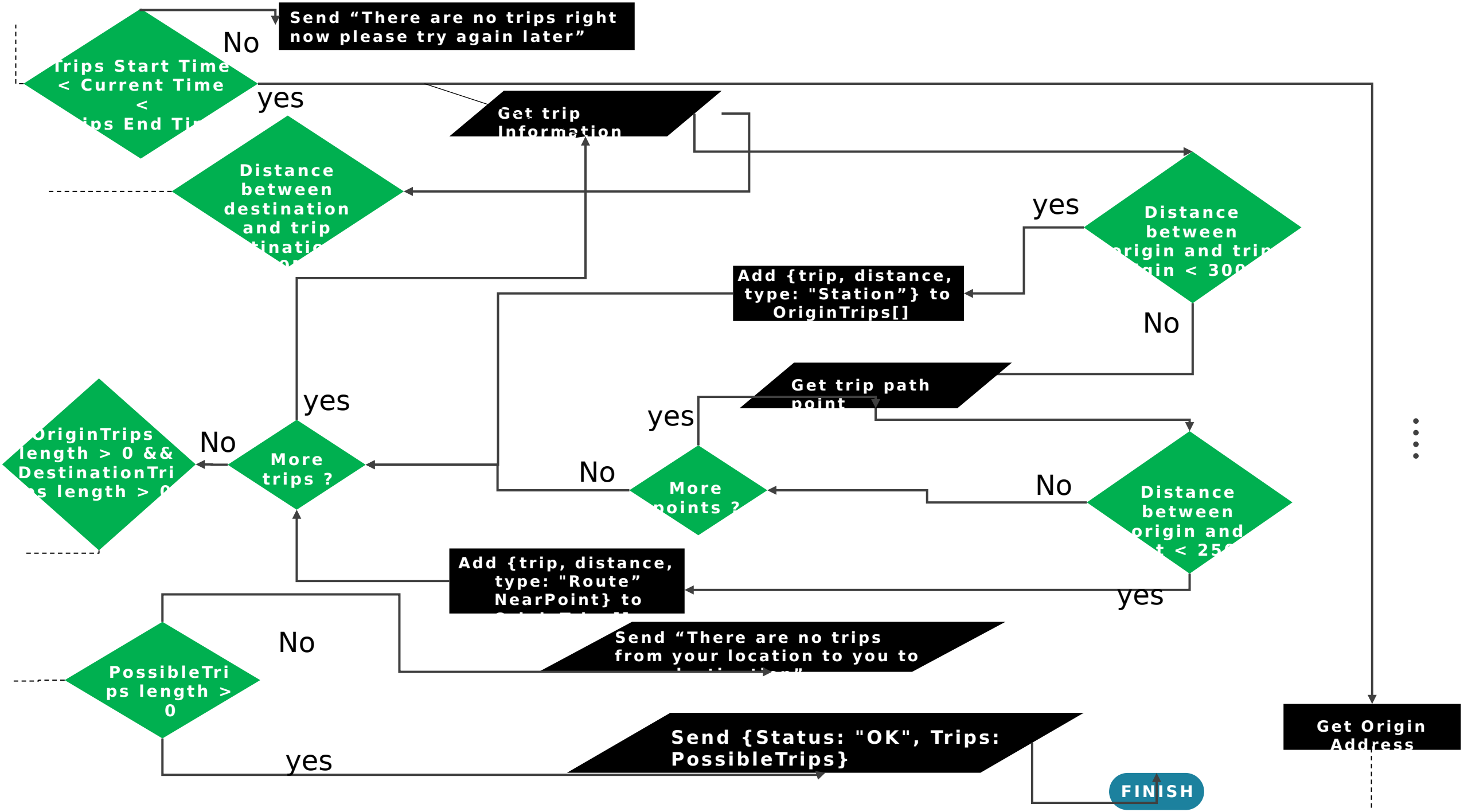
Role

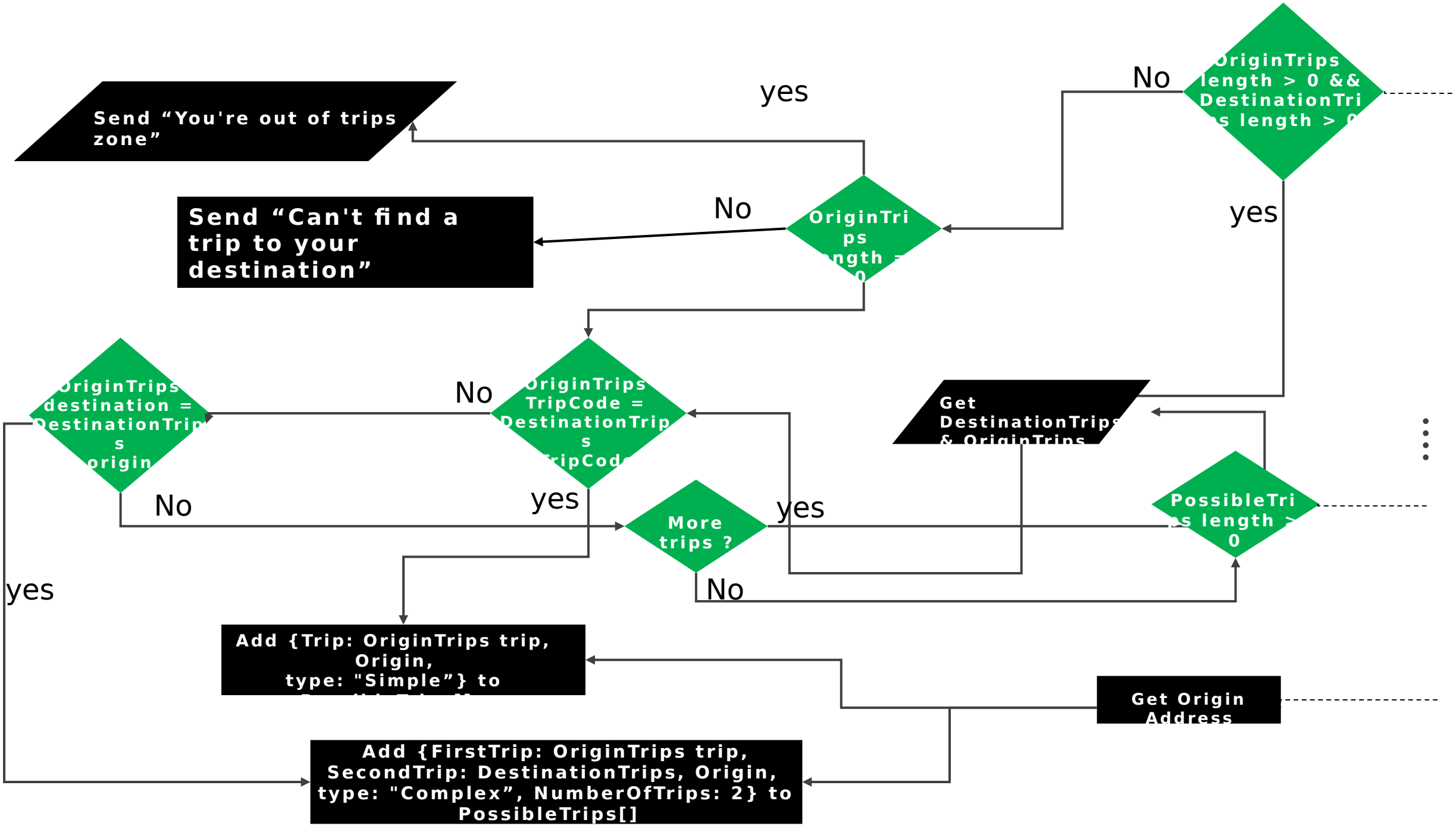
- Provides the mobile application “user” with all possible trips from the user’s location “origin” to a destination

Development tools

- Firebase Cloud Functions
- Firestore
- Axios
- CORS


START
Origin = {Latitude, Longitude} Destination = {Latitude, Longitude}
Send “Neither the origin nor the destination could be equal to null”
No
Origin & Destination
yes
Trips Start Time
Trips End
Trips Start Time < Current Time <
No
Distance between destination and trip
yes
More points ?
No
yes
Get trip path point
More trips ?
No
Add {trip, distance, type: "Station”}
Distance between destination and point < 250M
No
yes
Add {trip, distance, type: "Route” NearPoint} to
OriginTrips length > 0 &&


Send “There are no trips right now please try again later”
No
Trips Start Time < Current Time < ips End Ti
yes
Get trip Information
Distance between destination and trip inatio
yes
Distance between rigin and trip gin < 300
Add {trip, distance, type: "Station”} to OriginTrips[]
No
Get trip path point
yes
yes
OriginTrips length > 0 && DestinationTri s length > 0
No
More trips ?
No
More points ?
No
Distance between origin and t < 250
Add {trip, distance, type: "Route” NearPoint} to
yes
No
Send “There are no trips from your location to you to
PossibleTrips length > 0
yes
Send {Status: "OK", Trips: PossibleTrips}
Get Origin Address
FINISH


Send “You're out of trips zone”
yes
No
OriginTrips length > 0 && DestinationTrips length > 0
Send “Can't find a trip to your destination”
No
OriginTrips length = 0
yes
OriginTrips destination = DestinationTrips origin
No
OriginTrips TripCode = DestinationTrips TripCode
Get DestinationTrips & OriginTrips
No
yes
More trips ?
yes
PossibleTrips length > 0
yes
No
Add {Trip: OriginTrips trip, Origin, type: "Simple”} to
Get Origin Address
Add {FirstTrip: OriginTrips trip, SecondTrip: DestinationTrips, Origin, type: "Complex”, NumberOfTrips: 2} to PossibleTrips[]

HardWare

Hardware Requirements

Microcontroller

Executes the code necessary to send data

GSM module

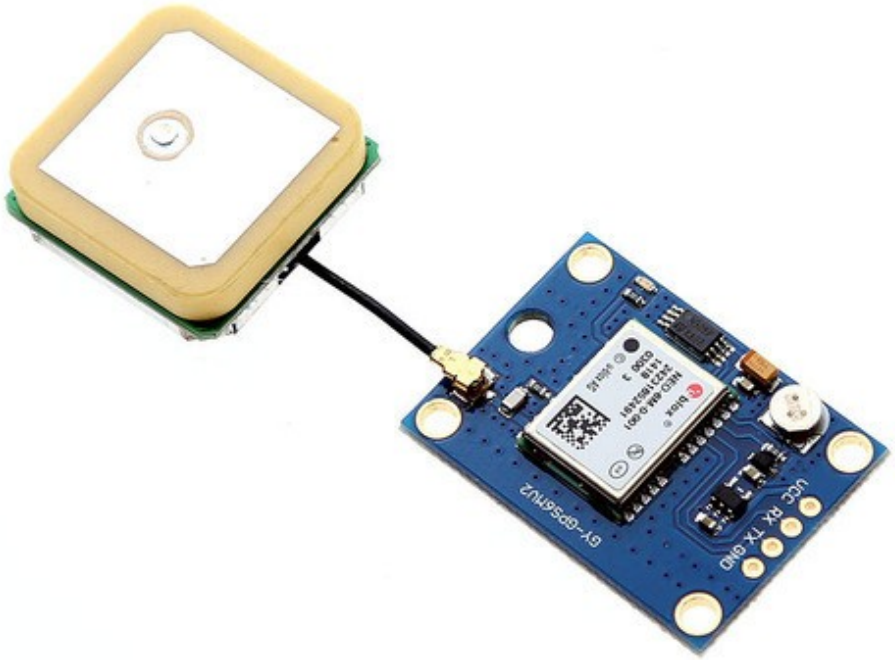
Connects the microcontroller with the internet

GPS module

Collects the location and speed data of the vehicle

GPS

- Get vehicle latitude, longitude, and speed in kmph
- TinyGPSPlus library



SIM800L

- GSM/GPRS module
- Update the vehicle location and active Trip via cloud function throw internet
- SIM800L HTTP connector library

Cloud Functions

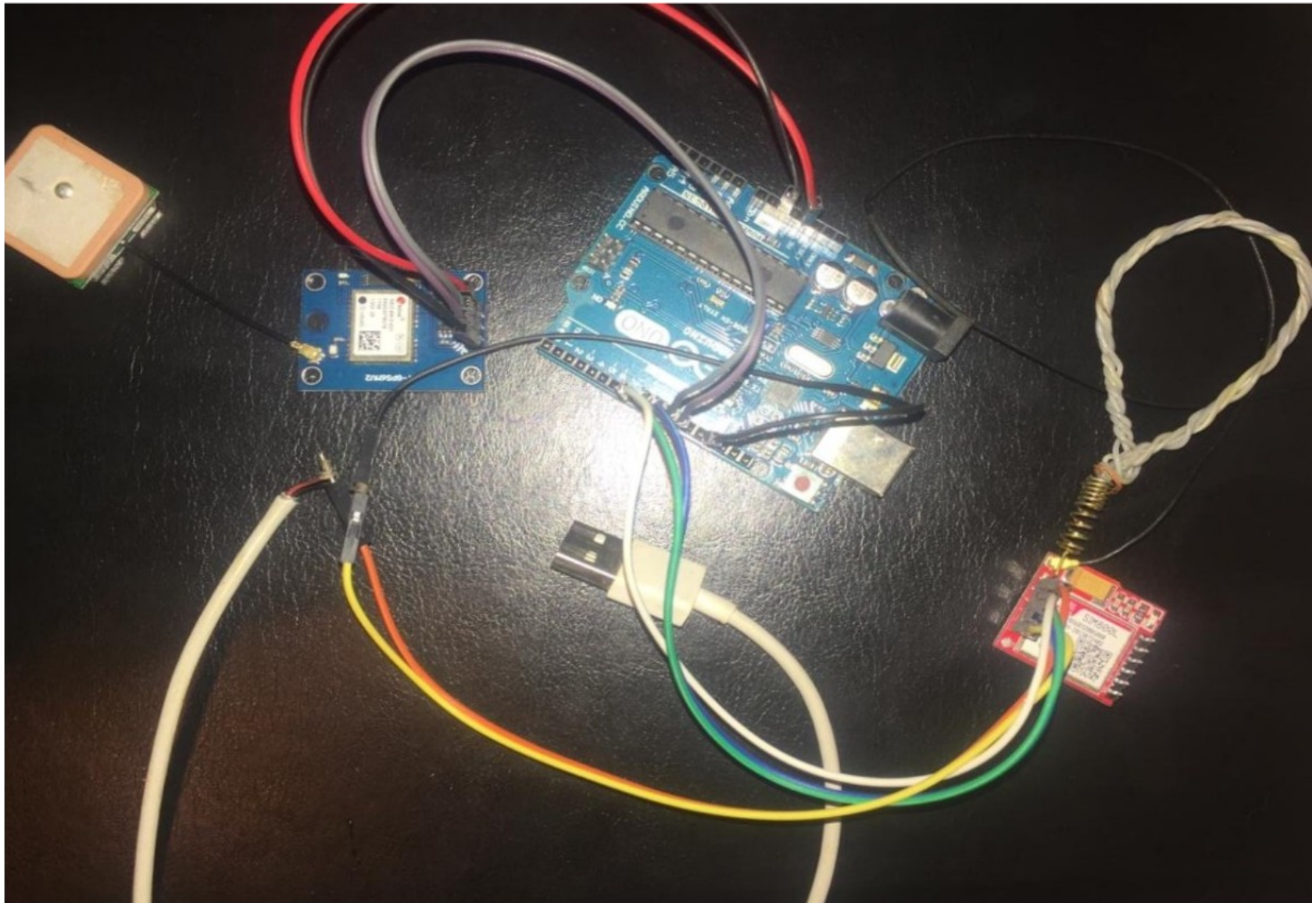
Active Trip Cloud Function

- The function determines which trip the vehicle is using
- It accepts the plate number, latitude and longitude
- It uses the spherical law of cosines to determine the trip

Location Cloud Function

- This function also accepts the plate number and location
- But it is repeatedly called to keep the database updated
- It also keeps a record of all vehicles locations with timestamps

Hardware Connections



Arduino Run-time

```
10:28:29.542 -> Initialize AT command...
10:28:30.555 -> Initialize Ok
10:28:31.600 -> Signal (Strenght: 0)...
10:28:32.648 -> Signal (Strenght: 0)...
10:28:32.682 -> Signal OK (Strenght: 17)
10:28:33.726 -> Network registration...
10:28:34.774 -> Network registration OK
10:28:35.954 -> GPRS APN config OK
10:28:36.021 -> GPRS USER config OK
10:28:36.089 -> GPRS PWD config OK
10:28:37.102 -> GPRS connecting...
10:28:37.640 -> GPRS connected
10:28:39.127 -> HTTP initialization...
10:28:39.768 -> HTTP initialization OK
10:28:43.282 -> Active Trip PAYLOAD: {"Lat":"31.1248","Long":"29.7765"}
10:28:44.262 -> Starting Active Trip HTTP POST...
10:28:51.643 -> Active Trip HTTP POST successful (4 bytes)
10:28:51.676 -> Received : DONE
10:28:55.644 -> PAYLOAD: {"Lat":"31.1248","Long":"29.7764","Speed":"0.41"}
10:28:56.660 -> Starting HTTP POST...
10:29:01.687 -> HTTP POST successful (2 bytes)
10:29:01.687 -> Received : OK
```

⋮

Admin Panel

Administrator Panel

Role

- System monitoring
- Insert/modify stations
- Insert/delete transportation trips
- Insert/modify vehicles with assigned trips

External APIs

- Google Maps JavaScript API
- Google Places API
- Google Directions API

Development tools

- Reactjs
- Bootstrap
- Mdbreact
- Firebase authentication
- Firestore
- http-proxy-middleware
- Axios

Administrator Panel

Why React.js

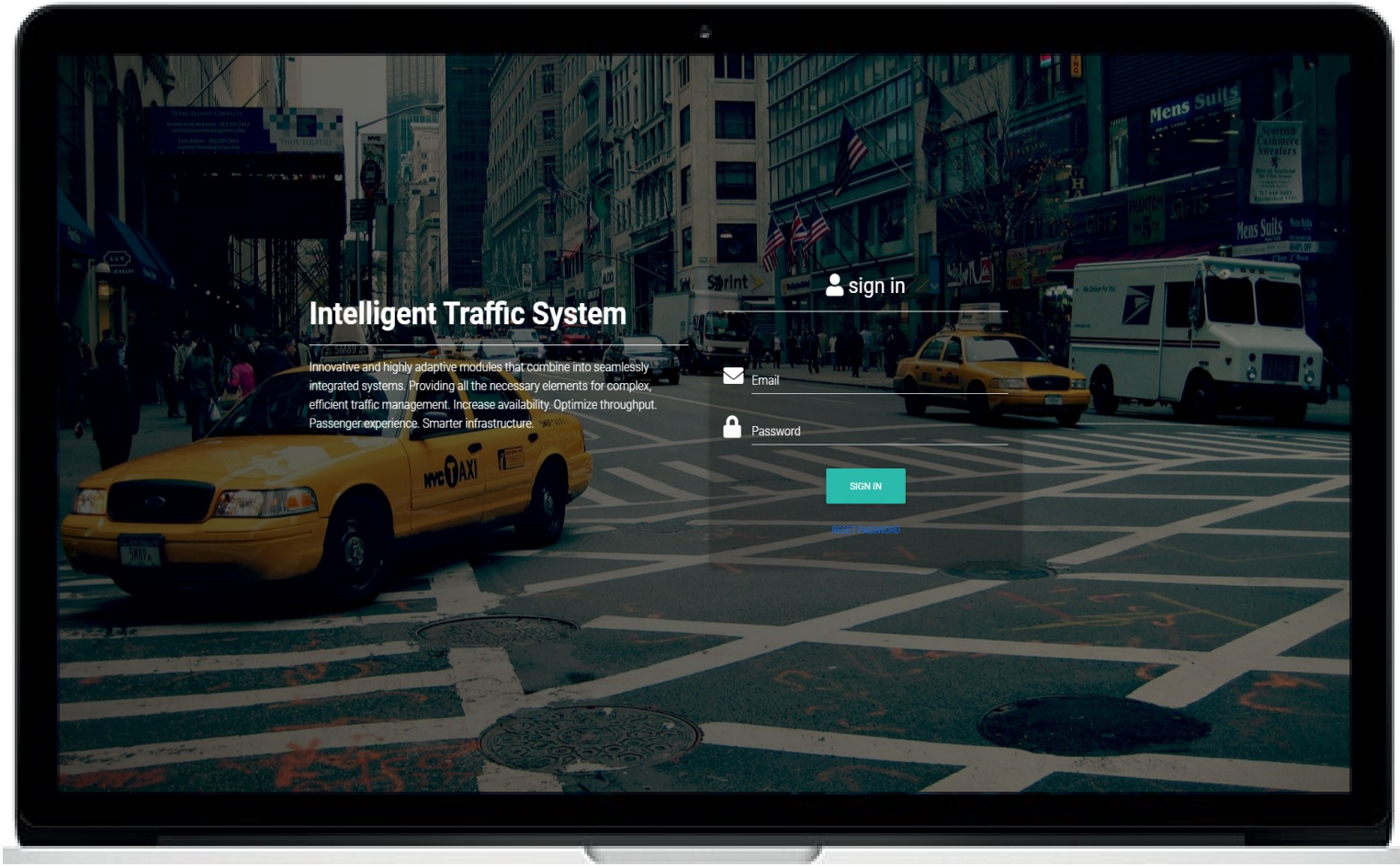
- Virtual Document Object-Model

Why Firestore

- Real-time NoSQL database that remain data synchronized

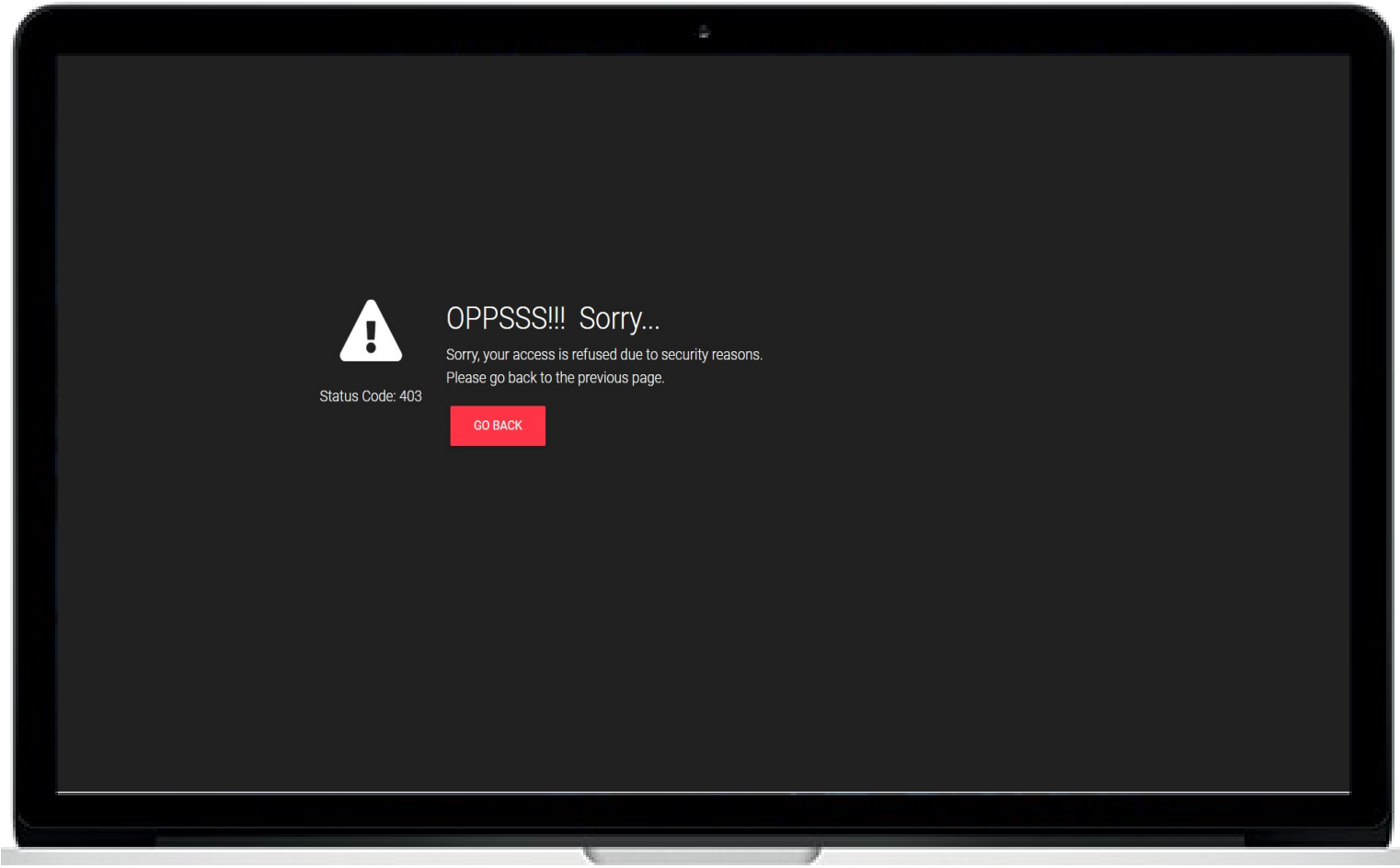
/login

Allows only super/administrator users to gain access to the application by entering their email and password



/403

If the user does not exist as super/administrator



/admin

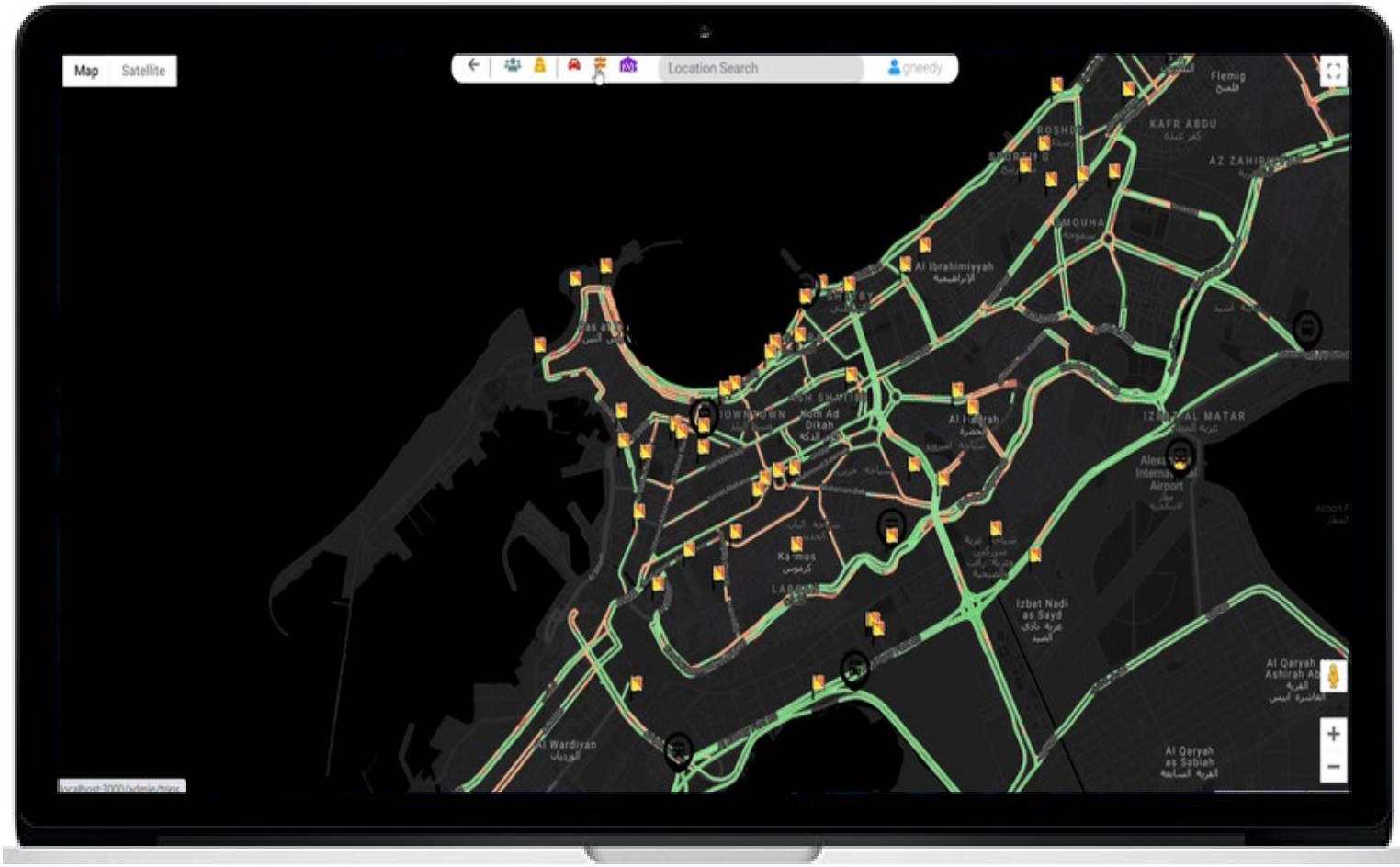
Monitor stations, accidents and vehicles motion on the map

/admin/stations

- Insert stations as a pin on the map with a certain location
- Modify stations information
- Monitor stations

/admin/trips

- Insert trips from a station to another
- Modify trips acceptance time period
- Monitor trips
- Delete trips



/admin/vehicles

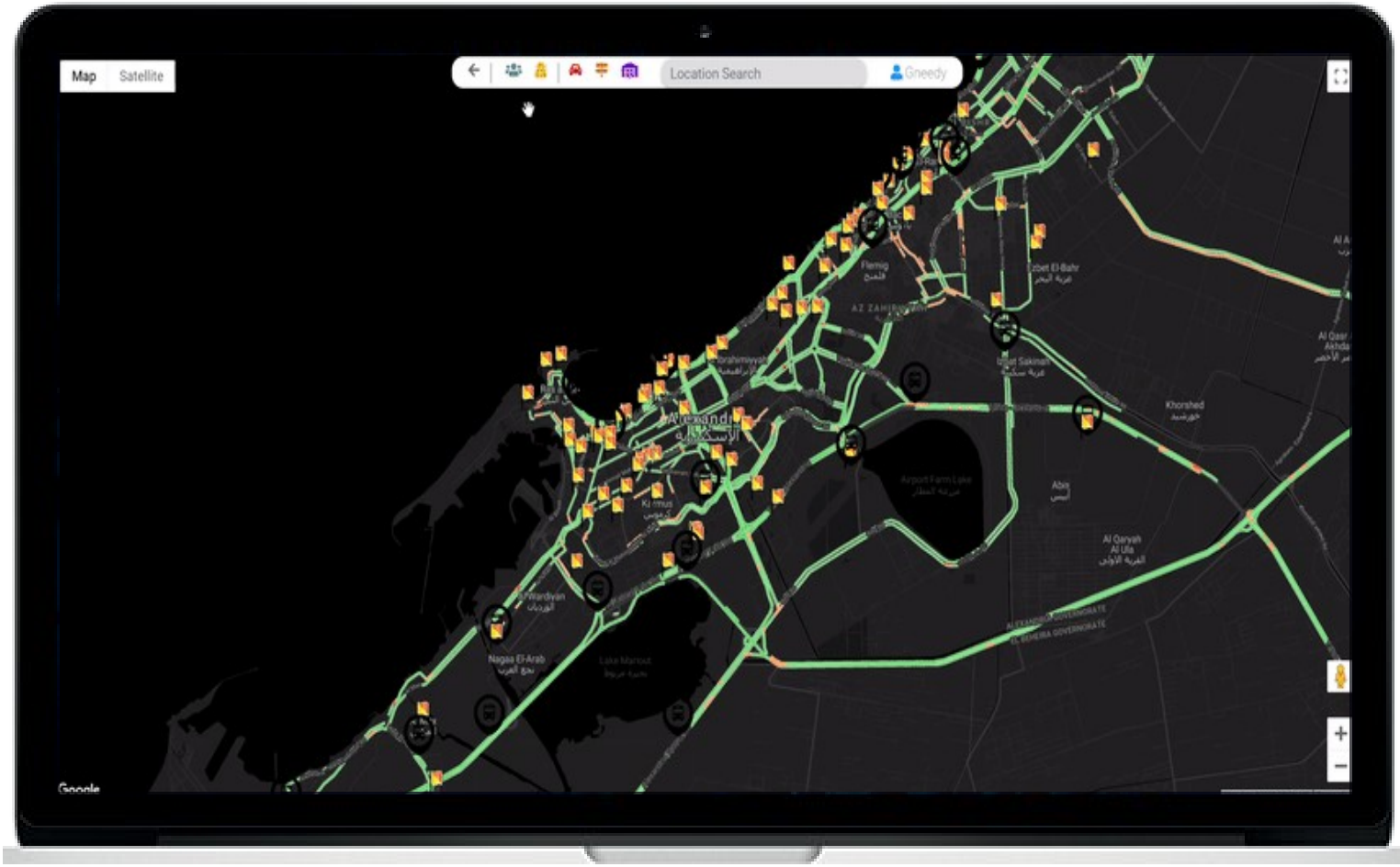
- Insert vehicles by plate number
- Assign trips for each vehicle
- Modify vehicles assigned trips
- Monitor vehicles and there assigned trips

/admin/users

Monitoring users in the system

/admin/ administrators

- Accessible by super administrators only
- Insert administrators and super administrators users
- Modify administrators users role
- Monitor administrators users



Mobile Application

React Native

React Native is a framework that allows you to build native mobile apps using JavaScript. Normally, you'd need to program your mobile app using Java (for Android) and Swift/Obj-C (for iOS).

React Native removes that requirement, leading to fully functional apps on both platforms in much less time and

Popular React Native apps

- Facebook
- Instagram
- Skype
- Discord

Advantages of React Native

Save time and money

If you need to develop an app for both iOS and Android, React Native is the best tool out there. It can reduce the codebase by about 95%, saving you time and money

Third-party plugins

React Native also allows you to easily incorporate third-party plug-ins and APIs, including maps and payment systems

Use existing skills

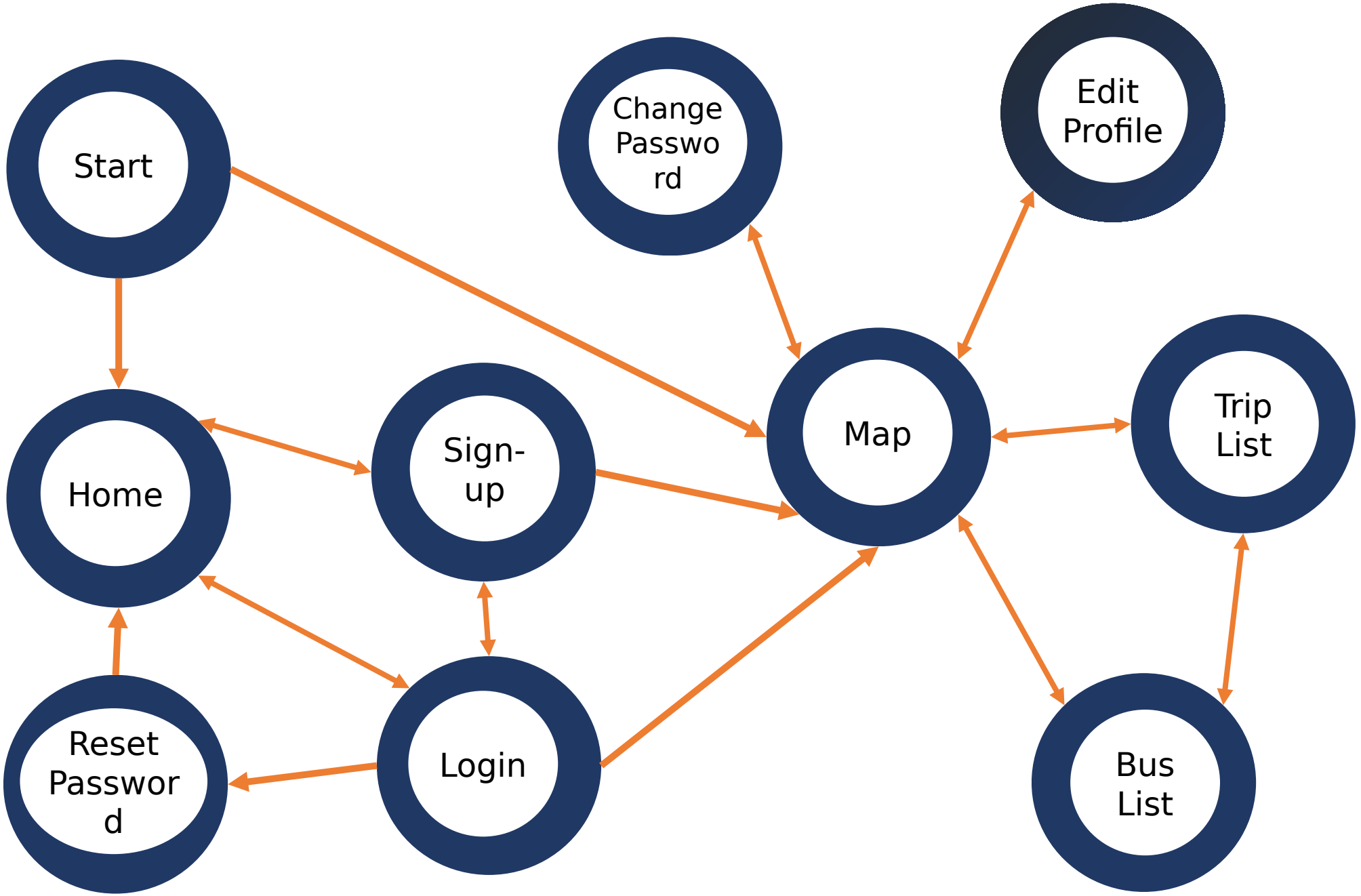
Since React Native is based on JavaScript, your developers will not need long to get to grips with it. Having said this, it’s an open-source framework, so if they ever need support, it’s widely available online.

Application Features

Easy to use

Monitori ng the bus

Screens interactions



Home Screen

Include :

- **Login Button :**
  navigate to Login screen

- **Sign-up Button :**
  navigate to Sign-up screen

Login Screen

Include :

- **Email Field**
- **Password Field**

Error Type

Front-end Error

- Email Required
- Invalid Email
- Password Required

Back-end Error

- Invalid Email
- Invalid Password

Sign-Up Screen

Include :

- **First Name Field**
- **Last Name Field**
- **Email Field**
- **Phone Field**
- **Password Field**

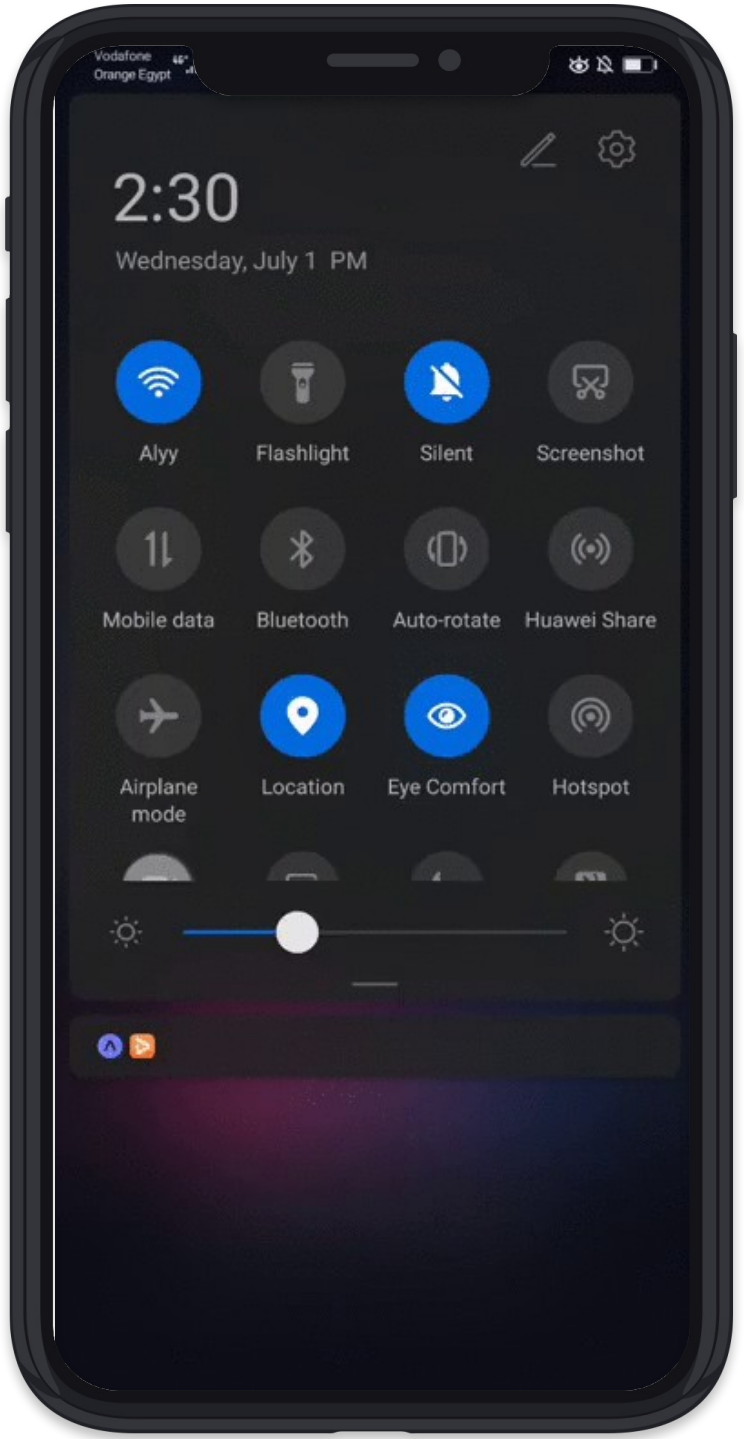
Error Type

Front-end Error

- Email Required / Invalid Email
- Phone Required (11 numbers)
- Password Required (min 6 char and num)
- First Name (min 3 characters)
- Last Name Required (min 3 characters)

Back-end Error

- E-mail Already in used



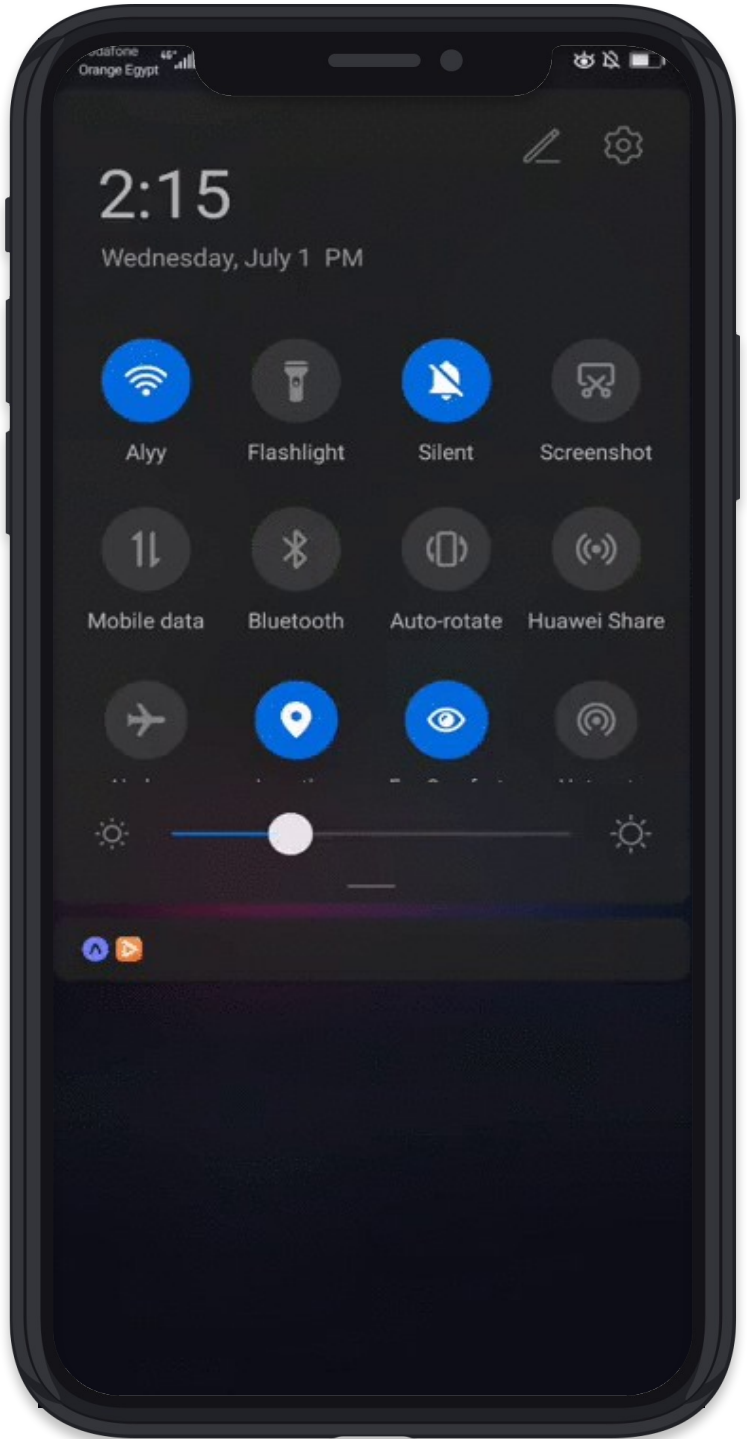
Reset Password Screen

Include :

- **Email Field**

Error Type

- **Front-end Error**
  - Email Required / Invalid Email
- **Back-end Error**
  - User with this email doesn't exist



Change Password and Profile Edit Screen

Include :

-----Edit----

- **First Name Field**
- **Last Name Field**

----Change Password----

- **Password Field**
- **Confirm Password Field**

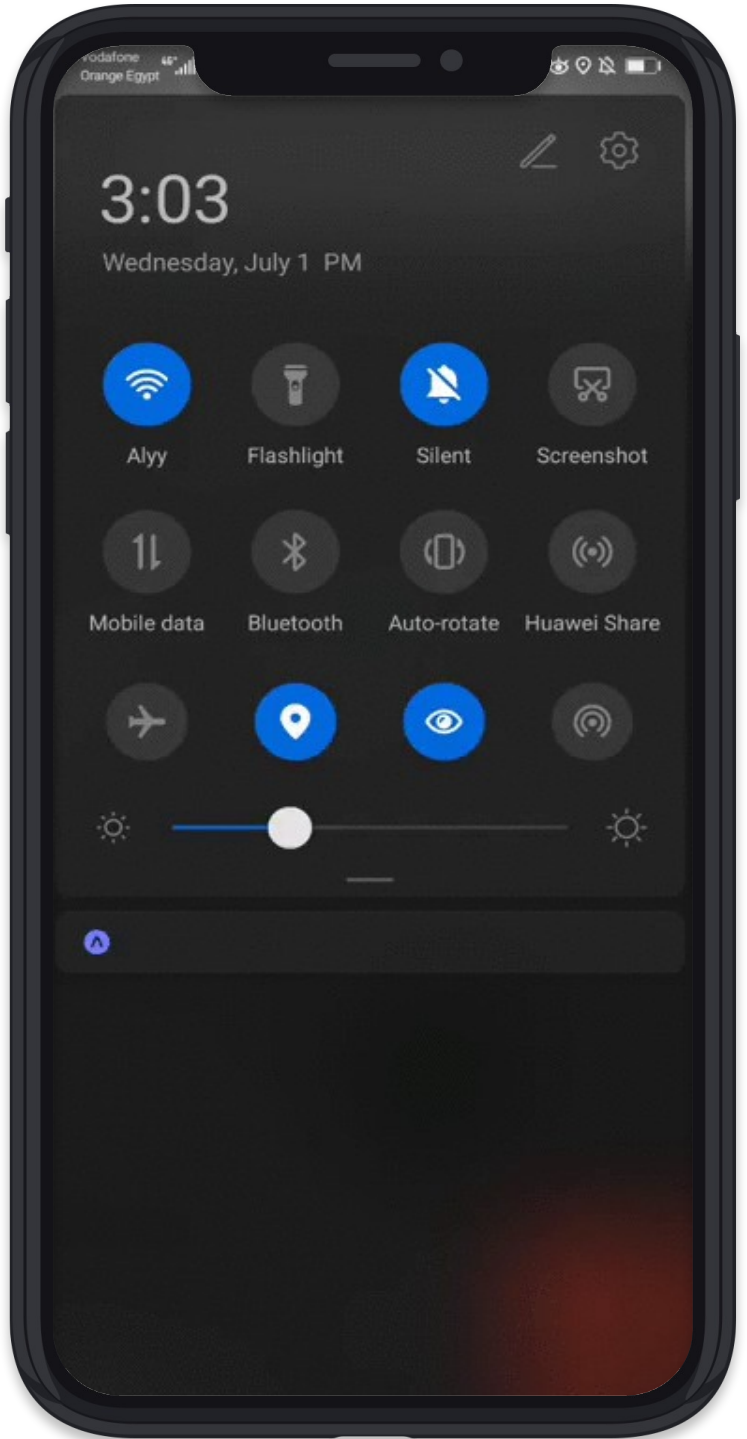
Error Type

Front-end Error

- Password must match
- Password Required
- Confirm Password Required

Back-end Error

- Password is too weak



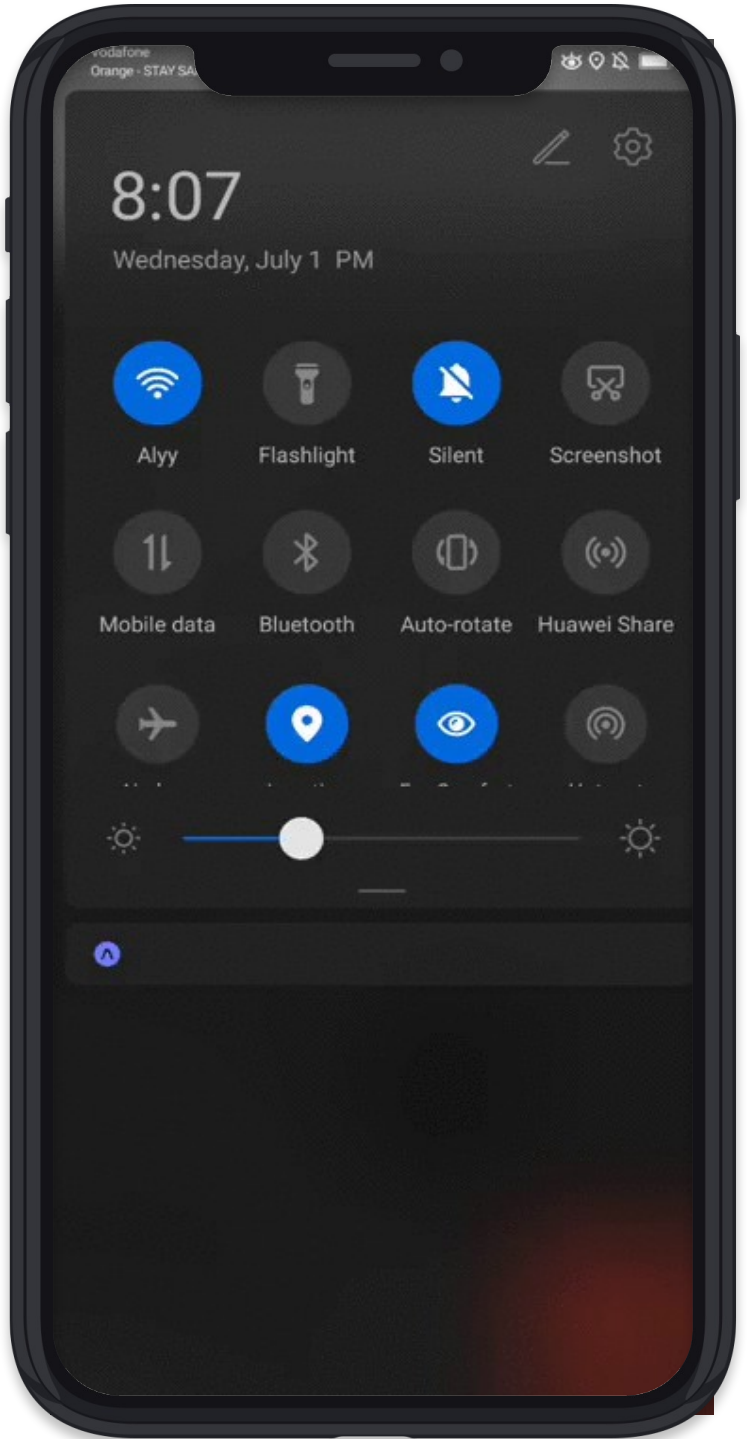
Map Screen

Include :

- Destination Search **:**
  Search about your destination
- **Confirm Button :**
  Choose your First Trip
- **Second Trip Button :**
  Choose your Second Trip
- **End Trip Button :**
  End your Trip
- **Trip List Consist of :**

One vehicle : Confirm Button / End Trip Button

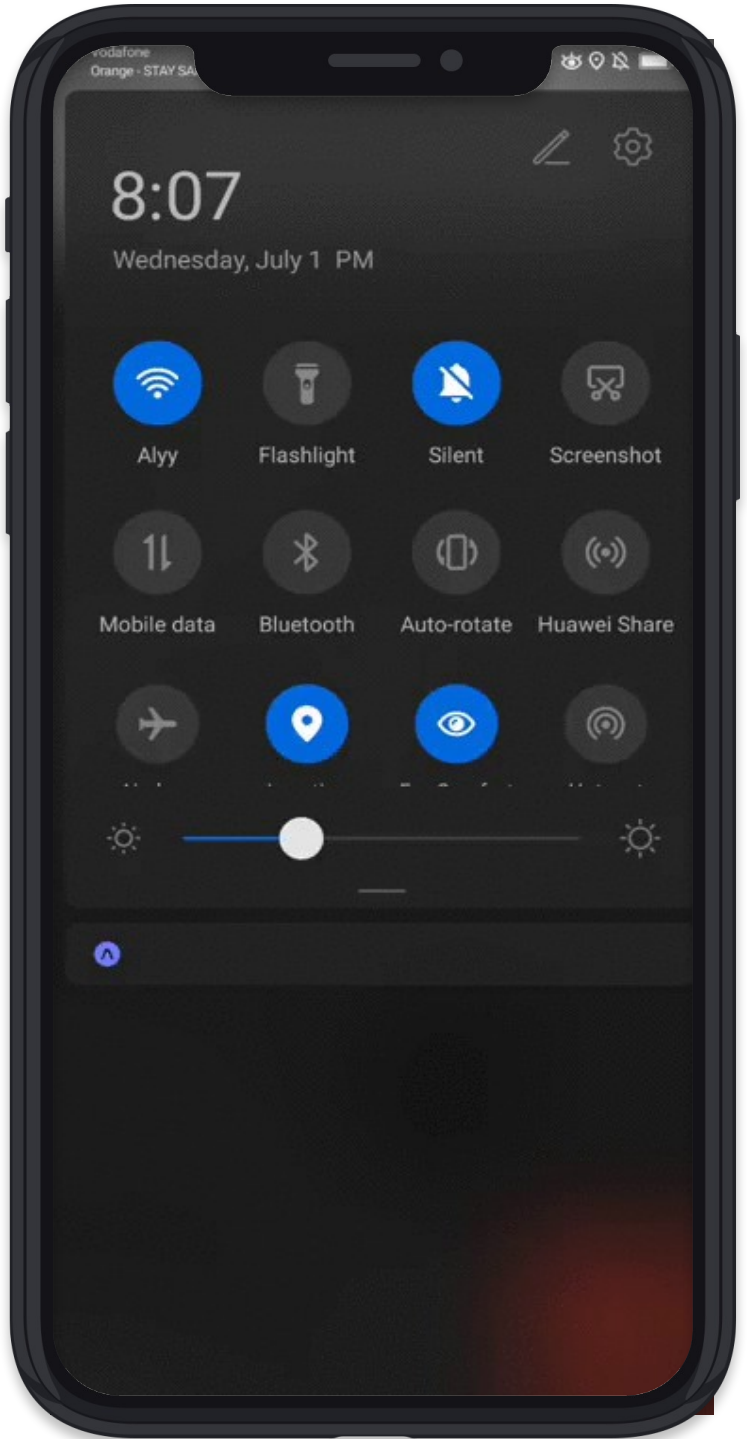
Two Vehicle :Confirm Button / Second Trip / End Trip Button



Map Screen

Include :

- Plus button **consist of :**
  - **Edit Profile**
  - **Change Password**
  - **Logout**
- **One vehicle :**
  reach your destination by one vehicle
- **Two vehicle :**
  reach your destination by two vehicle
- **Nearby point :**
  There is a nearby point shown in the

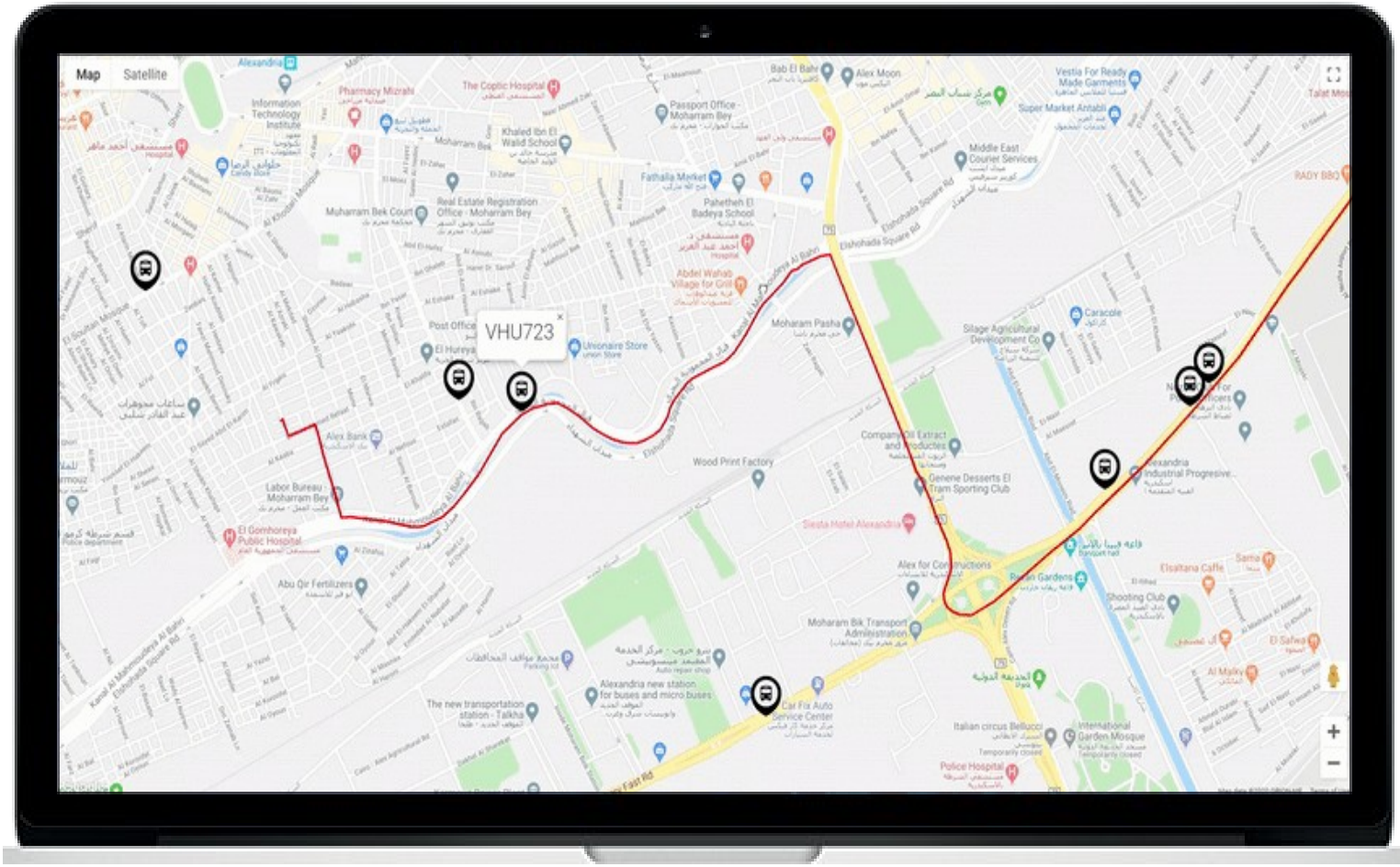


Simulation Test

/simulation

Accessible by super administrators only

Simulate vehicles motion



Thank You!